
GENERAL RELATIVITY
Tutorial problem set 7, 25.10.2019.

■ **PROBLEM 19** Riemann tensor.

Riemann tensor (with all indices down) is defined to be

$$R_{\lambda\sigma\mu\nu} = g_{\lambda\rho} R^{\rho}_{\sigma\mu\nu} = g_{\lambda\rho} \left(\partial_{\mu} \Gamma^{\rho}_{\nu\sigma} - \partial_{\nu} \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\alpha} \Gamma^{\alpha}_{\nu\sigma} - \Gamma^{\rho}_{\nu\alpha} \Gamma^{\alpha}_{\mu\sigma} \right). \quad (19.1)$$

- (a) Prove that Riemann tensor is antisymmetric in its first two indices,

$$R_{\lambda\sigma\mu\nu} = -R_{\sigma\lambda\mu\nu}, \quad (19.2)$$

and that it is antisymmetric in its last two indices,

$$R_{\lambda\sigma\mu\nu} = -R_{\lambda\sigma\nu\mu}. \quad (19.3)$$

Furthermore, prove that it is symmetric under the exchange of left-hand pair and right-hand pair of indices,

$$R_{\lambda\sigma\mu\nu} = R_{\mu\nu\lambda\sigma}. \quad (19.4)$$

Finally, prove the identity

$$R_{\lambda\sigma\mu\nu} + R_{\lambda\mu\nu\sigma} + R_{\lambda\nu\sigma\mu} = 0. \quad (19.5)$$

(The last identity is equivalent to writing $R_{\lambda[\sigma\mu\nu]}$, can you see why?)

- (b) Using the symmetries of Riemann tensor derived in (a) show that Riemann tensor in D dimensions has $D^2(D^2 - 1)/12$ independent components.
- (c) In $D = 2$ dimensions Riemann tensor has only one independent component. Convince yourself that we can write it in the following form

$$R_{\lambda\sigma\mu\nu} = f(x) \left(g_{\lambda\mu} g_{\sigma\nu} - g_{\lambda\nu} g_{\sigma\mu} \right), \quad (19.6)$$

where $f(x)$ is a scalar function of coordinates. Determine what f is by requiring that the appropriate contractions of the Riemann tensor yield the Ricci scalar.

- (d) Show the Bianchi identities,

$$\nabla_{\alpha} R_{\rho\sigma\mu\nu} + \nabla_{\mu} R_{\rho\sigma\nu\alpha} + \nabla_{\nu} R_{\rho\sigma\alpha\mu} = 0, \quad (19.7)$$

$$\nabla^{\alpha} R_{\alpha\beta} - \frac{1}{2} \nabla_{\beta} R = 0. \quad (19.8)$$

■ **PROBLEM 20** Geodesic deviation.

Consider a pair of nearby freely falling particles that travel on trajectories $x^\mu(\tau)$ and $x^\mu(\tau) + \delta x^\mu(\tau)$. Their equations of motion are

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu(x) \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0 , \quad (20.1)$$

$$\frac{d^2(x^\mu + \delta x^\mu)}{d\tau^2} + \Gamma_{\alpha\beta}^\mu(x + \delta x) \frac{d(x^\alpha + \delta x^\alpha)}{d\tau} \frac{d(x^\beta + \delta x^\beta)}{d\tau} = 0 . \quad (20.2)$$

(a) Show that the difference between these two equations to first order in δx^μ is

$$\frac{d^2(\delta x^\mu)}{d\tau^2} + \frac{\partial \Gamma_{\alpha\beta}^\mu}{\partial x^\nu} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \delta x^\nu + 2\Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{d(\delta x^\beta)}{d\tau} = 0 . \quad (20.3)$$

(b) Show that this can be rewritten in terms of covariant derivatives along the curve $x^\mu(\tau)$,

$$\frac{D^2(\delta x^\mu)}{d\tau^2} = -R^\mu{}_{\alpha\nu\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \delta x^\nu . \quad (20.4)$$

Remember the definition of the directional derivative along a curve $x^\alpha(\tau)$,

$$\frac{D}{d\tau} \equiv \frac{dx^\alpha}{d\tau} \nabla_\alpha . \quad (20.5)$$

■ **PROBLEM 21** Weyl tensor.

The Weyl tensor in D dimensions is defined to be

$$C_{\rho\sigma\mu\nu} = R_{\rho\sigma\mu\nu} - \frac{2}{(D-2)} \left(g_{\rho[\mu} R_{\nu]\sigma} - g_{\sigma[\mu} R_{\nu]\rho} \right) + \frac{2}{(D-1)(D-2)} g_{\rho[\mu} g_{\nu]\sigma} R . \quad (21.1)$$

(a) Determine the number of independent components of the Weyl tensor in D dimensions.

Hint: Show that it has the same symmetries as the Riemann tensor, and in addition $C^\alpha{}_{\mu\alpha\nu} = 0$.

(b) Show that for $D \geq 3$ the Weyl tensor satisfies the following version of Bianchi identity,

$$\nabla_\rho C^\rho{}_{\sigma\mu\nu} = 2 \frac{D-3}{D-2} \left(\nabla_{[\mu} R_{\nu]\sigma} + \frac{1}{2(D-1)} g_{\sigma[\mu} \nabla_{\nu]} R \right) . \quad (21.2)$$

Hint: Use the Bianchi identities from problem 1.

(c) Can you argue why the expression above is not defined for $D < 3$?